

Charlie Boswell, Embedded Systems Engineer

Employment History

Electronics and Computing Degree Apprentice in Advanced Control Unit (2020 - present)

JET CODAS / ACU projects

- Worked with VxWorks real-time operating system while diagnosing a fault in KG1-RT interferometer data acquisition
- Learned modern dev-ops practices while contributing to development of RTCC2 GUI
- Produced onboarding materials for MARTe2 learners and created forum for external and internal users
- Produced documentation e.g. MAST-U PCS upgrade documentation

MAST-U projects

- Led configuration and deployment of new D-TACQ data acquisition units in MAST-U PCS upgrade
- Developing source table controller FPGA for double beam box power supplies

LiBrTI Work

- I work as a part of the EC&I team, planning SCADA style, PLC driven control systems
- I am the control lead for a pincell test rig, controlling temperature in a pressure vessel.

Innovation Projects

Assisted with several innovation projects. I designed and conducted experiments, collected data and kept documentation. Projects include Pi Core Autonomy, Soft sensors for robotics, Digital nuclear track detectors and Nimbus fast neutron diagnostic.

Other Achievements

- Led over 100 tours of our facility for stakeholder, education and public groups. Helped at stalls at science festivals including New Scientist Live
- Mentored three GCSE work experience students directly for week long placements
- Lead a control system community group, organising events for around 30 engineers

Teacher of English as a second language at Northeast Yucai, China (2017 - 2019)

- Taught classes of forty students, aged between eleven and fourteen with side classes for all ages

- Created and delivered my own lesson plans, in English
- Built public speaking confidence and technique
- Created lessons that engaged even reluctant students
- Improving mindful communication, awareness of audience

Education

BEng Electronics and Computing, University of West of England (2020 - 2025)

- Dissertation title: FPGA Partial Reconfiguration for Universal Hardware Interfaces
- Acquired 1st class degree electronics and computing
- Average module score: 80.9
- Communications engineering optional module. Studied analogue and digital communications including information theory, modulation techniques, stochastic noise, error detection
- Core modules in analogue circuits, control, embedded systems, power electronics, signal processing, systems engineering, FPGA development

Apprenticeship (2020 - 2025)

Completion of the apprenticeship is a separate award granted by IET. It required proof that I work to the standard of an incorporated engineer.

BSc Music Technology, Staffordshire University (2012 - 2015)

Sparked an interest in electronics through modules about microphone and speaker design. When later studying electronics I discovered that signal processing techniques I was learning are applied in audio too.

Volunteering

Assisted reading with ARCh (2024 - present)

I work with young children with reading difficulties to inspire a love of reading and to provide a space for one-on-one literacy practice.

In2Science mentor (2025 - present)

In2Science is a charity helping disadvantaged groups explore careers in science. I mentor an A-level student via video calls to advise skills to prioritise and potential career paths.